(a) The Peyer's patch is a representative of the extensive MALT system that is found in the intestine. (b) A stained tissue cross-section of Peyer's patch lymphoid nodules in the intestinal submucosa is schematically diagrammed in (c). The intestinal lamina propria contains loose clusters of lymphoid cells and diffuse follicles.

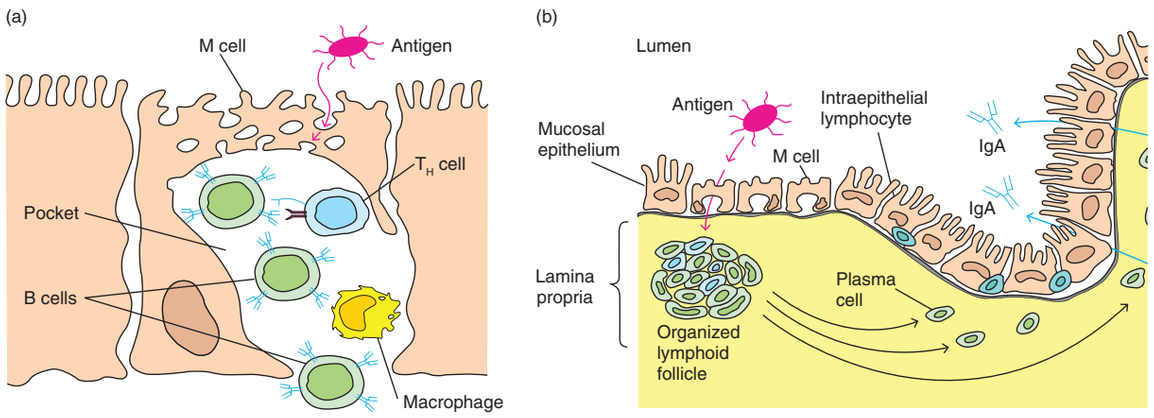


Figure 13.8: Structure of M cells and production of IgA:

(a) M cells, situated in mucous membranes, endocytose antigen from the lumen of the digestive, respiratory, and urogenital tracts. The antigen is transported across the cell and released into the large basolateral pocket. (b) Antigen transported across the epithelial layer by M cells at an inductive site activates B cells in the underlying lymphoid follicles. The activated B cells differentiate into IgA-producing plasma cells, which migrate along the lamina propria, the layer under the mucosa. The outer mucosal epithelial layer contains intraepithelial lymphocytes, of which many are T cells.

types of MALT. The system most studied is the gut associated lymphoid tissue (GALT). GALT include the tonsils, adenoids, and appendix and specialized structures called peyer's patches (Figure 13.7) in the small intestine, which collect antigen from the epithelial surfaces of the gastrointestinal tract. In peyer's patches, the antigen is collected by specialized epithelial cells called M cells (Figure 13.8). The lymphocytes form a follicle consisting of a large central dome of B lymphocytes surrounded by small numbers of T lymphocytes. Similar but more diffusely organized aggregates of lymphocytes protect the respiratory epithelium, where they are known as bronchial- associated lymphoid tissue (BALT).

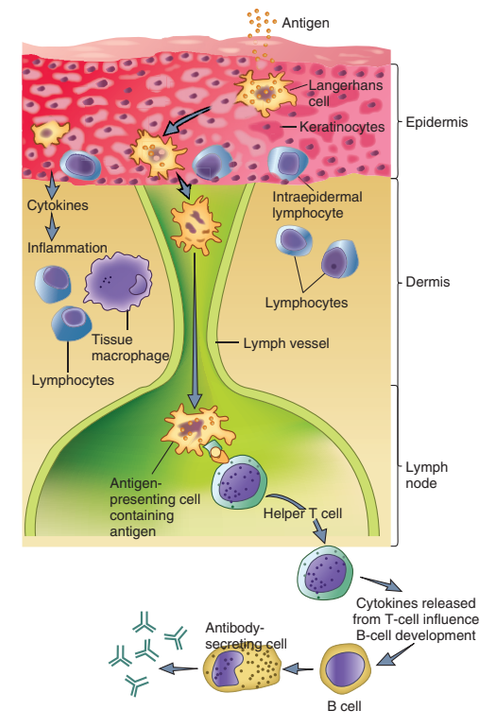


Figure 13.9: SALT

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